

Executive KPI Dashboard

Case Study & KPI Reference Guide

How a board-ready SaaS dashboard turns raw data into decisions

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1. The Business Problem

Software as a Service (SaaS) founders and product leaders face a common operational challenge: their most important business metrics live in completely different places. Monthly Recurring Revenue (MRR) is in Stripe. Customer churn data is in a Customer Relationship Management (CRM) system. Customer Acquisition Cost (CAC) is in a marketing spreadsheet. Lifetime Value (LTV) is calculated by different people using different assumptions and rarely agrees between departments.

When a founder needs to prepare for a board meeting or investor conversation, pulling these numbers together takes days. By the time the presentation is ready, some figures are already out of date. Benchmark comparisons against industry standards have to be done manually.

This dashboard solves that problem. It aggregates all of these signals into a single, board-ready view that updates in real time and generates a one-page KPI snapshot in one click.

2. What the Dashboard Does

The Executive KPI Dashboard is a three-tab interactive application that gives SaaS founders, executives, and investors a complete picture of business health without any manual data assembly.

Tab 1 - Executive Summary

Nine Key Performance Indicator (KPI) cards arranged in three rows: Revenue, Unit Economics, and Retention. Each card shows the current value, a month-on-month (MoM) delta, and a Red/Amber/Green (RAG) benchmark status with colour-coded borders. Green = above industry benchmark. Amber = acceptable but requires monitoring. Red = below benchmark and requires action.

A plain-English insights panel below the cards explains what the numbers mean in business terms - automatically generated from the data. A one-click board PDF export button generates a formatted snapshot for investor meetings.

Tab 2 - Growth

Three charts that tell the acquisition and revenue story: an MRR trend area chart, a new customers bar chart, and an MRR bridge waterfall chart. The waterfall is the most analytically powerful - it breaks current MRR into Prior MRR + New MRR + Expansion MRR - Churned MRR = Current MRR, answering the board question: are we growing from new sales, from existing customers expanding, or are we fighting churn?

Tab 3 - Efficiency

Four charts focused on unit economics: LTV to CAC (LTV:CAC) ratio trend with benchmark lines, CAC payback period trend with threshold lines, a Rule of 40 gauge (speedometer style with Red/Amber/Green zones), and a monthly churn rate area chart with target and threshold benchmark lines.

Segment Filtering

All three tabs respond to sidebar filters for plan tier (Starter / Growth / Enterprise) and region (UK / US / EU / Asia-Pacific). Selecting a segment re-aggregates all KPIs from the underlying customer-level data in real time. This means a founder can instantly answer 'what is our LTV:CAC for Enterprise customers in the US?' without any manual data manipulation.

3. The Metrics Story - What Each KPI Means

Revenue Metrics

MRR - Monthly Recurring Revenue

Benchmark: Growing >5% month-on-month

The heartbeat metric of any subscription business. MRR is the total predictable revenue collected from all active customers in a single month. It excludes one-off payments and only counts recurring subscription fees. If MRR is growing consistently, the business is healthy. If it plateaus or declines, it is a signal that acquisition is not keeping pace with churn.

ARR - Annual Recurring Revenue

Benchmark: Consistent growth trajectory

MRR multiplied by 12. ARR is the annualised view of the revenue base and is the number that investors, acquirers, and boards use to size a SaaS business. A business with GBP250k MRR has an ARR of GBP3M.

NRR - Net Revenue Retention

Benchmark: >=110% best-in-class, >=100% acceptable, <100% at risk

NRR measures what happens to revenue from existing customers over time. Formula: (starting MRR + expansion MRR - churned MRR) / starting MRR. An NRR above 100% means the existing customer base is growing the business on its own, even without a single new sale. Best-in-class SaaS companies (like Snowflake and Datadog) sustain NRR above 130%. An NRR below 100% means churn is outpacing expansion - the business has a leaky bucket problem.

Unit Economics Metrics

LTV:CAC - Lifetime Value to Customer Acquisition Cost Ratio

Benchmark: >=3x healthy, 2-3x monitor, <2x unsustainable

The most important unit economics metric. LTV is the total revenue a business can expect from one customer across their entire relationship (calculated as Average Revenue Per User (ARPU) / monthly churn rate). CAC is what it cost to acquire that customer. An LTV:CAC of 3x means every GBP1 spent on acquisition returns GBP3 in lifetime revenue. Below 2x, the business is spending more than it earns back - acquisition is destroying value.

CAC Payback - Customer Acquisition Cost Payback Period

Benchmark: <12 months Green, 12-18 months Amber, >18 months Red

The number of months required to recover the cost of acquiring a customer from that customer's revenue. A CAC payback above 18 months is a cash-flow warning: the business is funding a long gap between acquisition spend and return. Venture-backed SaaS companies typically target under 12 months. Enterprise SaaS can accept longer payback periods due to contract length.

Rule of 40 - Growth Rate + Profit Margin Score

Benchmark: >=40 Green, 20-40 Amber, <20 Red

A composite benchmark widely used by venture capitalists and growth-stage boards. The score is: year-on-year MRR growth percentage + estimated profit margin percentage. A score of 40 or above means the business is balancing growth and profitability well. Below 40 means one of the two levers needs attention. This dashboard uses year-on-year growth (not annualised month-on-month) to prevent inflated scores during early rapid growth.

Retention Metrics

Churn Rate - Monthly Customer Churn Rate

Benchmark: <=3% best-in-class, <=5% acceptable (Small and Medium Business SaaS)

The percentage of paying customers who cancelled their subscription in a given month. Monthly churn of 5% compounds to roughly

46% annual churn - meaning the business must replace nearly half its customer base every year just to stay flat. Even a 1% improvement in monthly churn has a dramatic compounding effect on revenue over 12-24 months.

NPS - Net Promoter Score

Benchmark: No universal benchmark (varies by industry)

Customers are asked: 'How likely are you to recommend us?' on a 0-10 scale. Promoters (9-10) minus Detractors (0-6) = NPS. Ranges from -100 to +100. NPS is tracked as an early warning signal for future churn - customers who are unlikely to recommend are more likely to cancel. The dashboard displays NPS without a RAG benchmark because standards vary too much between industries to set a universal threshold.

Active Customers - Total Active Paying Customers

Benchmark: Consistent growth month-on-month

The total number of customers with an active subscription at period end. This is the denominator behind every percentage metric on the dashboard. A rising active customer count combined with a rising churn rate is a warning sign: new acquisition is masking retention problems that will compound later.

4. How the Dashboard Supports Decisions

Board and investor preparation

The Executive Summary tab (Tab 1) is designed so a founder can open the laptop in a meeting and have the conversation immediately. All nine KPIs are visible on one screen. RAG colours communicate status without the audience needing to know the benchmarks. The board PDF export produces a one-page snapshot that can be sent in advance of the meeting with no manual formatting.

Diagnosing growth quality

The MRR bridge chart (Tab 2) answers the question investors ask most: 'Where is the growth coming from?' If most of the bridge is New MRR, growth is acquisition-dependent and will slow if sales hiring slows. If Expansion MRR is a large proportion, the business has strong product-market fit and a natural growth engine in its existing base. If Churn MRR is large relative to New MRR, the business has a retention problem that acquisition cannot permanently hide.

Identifying unit economics problems early

The Efficiency tab (Tab 3) surfaces deteriorating unit economics before they appear in MRR. A rising CAC payback period combined with a flat or declining LTV:CAC ratio is a leading indicator that acquisition efficiency is dropping - possibly due to rising Customer Acquisition Cost as easy channels saturate, or due to lower-quality customer cohorts with higher churn. Catching this at month 6 is far less costly than discovering it at month 18.

Segment-level decisions

Applying a segment filter (e.g. Enterprise plan, US region) allows a founder to verify strategic assumptions. If the Enterprise segment shows LTV:CAC of 15x versus 3x for Starter, the data supports a decision to shift sales focus upmarket. If UK churn is 2% while APAC churn is 8%, the data surfaces a regional customer success problem that needs investigation.

5. Technical Architecture

Data pipeline

The pipeline follows the Extract, Transform, Load (ETL) pattern:

Extract: `pipeline/generate_data.py` creates 1,000 synthetic customer records (`data/customers.csv`). Each row represents one customer with signup month, plan tier, region, Average Revenue Per User (ARPU), CAC, churn month, Net Promoter Score (NPS), and expansion rate.

Transform: `pipeline/etl.py` loops through each of the 36 calendar months, identifies active/new/churned customers, calculates all KPIs, applies RAG benchmark flags, and generates plain-English insight text.

Load: outputs `data/monthly_kpis.csv` - 36 rows, 24 columns.

Segment filtering architecture

When a segment filter is applied in the dashboard, the `build_segment_kpis(plan, region)` function re-aggregates all 36 months of KPIs from the 1,000 customer records directly. This means the segment data is always consistent with the customer-level source rather than relying on pre-built pivot tables. Streamlit's `@st.cache_data` decorator caches results per segment combination, so the second selection within a session is instant.

Stack

Python 3.12 | pandas 2.2+ | Streamlit 1.35+ | Plotly 5.22+ | fpdf2 2.7+ | NumPy 1.26+

6. December 2024 Results Snapshot (demo data)

The demo dataset covers 36 months (January 2022 to December 2024) across 1,000 synthetic customers distributed across three plan tiers (Starter 50%, Growth 35%, Enterprise 15%) and four regions (UK 30%, US 35%, EU 25%, APAC 10%).

MRR (Monthly Recurring Revenue) - GBP250,143 [Amber]

MoM growth of 0.8% - stable but below the >5% high-growth benchmark.

ARR (Annual Recurring Revenue) - GBP3,001,722 [Amber]

Crosses the GBP3M ARR milestone in December 2024.

NRR (Net Revenue Retention) - 102.8% [Amber]

Revenue is retained from existing customers but expansion is below the best-in-class 110% threshold.

LTV:CAC (Lifetime Value to CAC Ratio) - 11.1x [Green]

Well above the 3x benchmark - healthy unit economics throughout the dataset.

CAC Payback (Payback Period) - 2 months [Green]

Exceptionally fast payback - acquisition spend recovers in 2 months.

Rule of 40 (Growth + Margin Score) - 78 [Green]

Strong score. Benchmark is 40. Indicates good balance of growth and efficiency.

Churn Rate (Monthly Customer Churn) - 4.1% [Amber]

Above the 3% best-in-class benchmark but within the 5% acceptable threshold. Worth monitoring.

NPS (Net Promoter Score) - 43 [N/A]

Score of 43 indicates satisfied customers. No RAG benchmark applied.

Active Customers (Paying Customers) - 470 [N/A]

Grown from 34 in January 2022 to 470 in December 2024.